

Quiz 2: Multiple Quantum Systems

Question 1. Consider the 2-party state

$$\rho_{AB} = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|).$$

Which of the following describe ρ_{AB} ?

- (A) Pure.
- (B) Separable.
- (C) Mixed.
- (D) Entangled.

Question 2. Consider the 2-party state

$$\rho_{AB} = \frac{1}{2}(|00\rangle + |11\rangle)(\langle 00| + \langle 11|).$$

Which of the following describe ρ_{AB} ?

- (A) Pure.
- (B) Entangled.
- (C) Mixed.
- (D) Purification of $\frac{1}{2}I_A$.

Question 3. Let $\tau_{AB} = |\psi\rangle\langle\psi|$, where

$$|\psi\rangle = \frac{1}{\sqrt{3}}(|01\rangle_{AB} - |00\rangle_{AB} + |11\rangle_{AB}).$$

What is $\tau_A = \text{tr}_B[\tau_{AB}]$?

$$\begin{aligned} A &= |1\rangle\langle 1|, & B &= \frac{1}{3}(2|0\rangle\langle 0| + |1\rangle\langle 1| + |0\rangle\langle 1| + |1\rangle\langle 0|), \\ C &= |1\rangle\langle 1| + |0\rangle\langle 1| + |1\rangle\langle 0|, & D &= \frac{1}{3}(|0\rangle\langle 0| + 2|1\rangle\langle 1| - |0\rangle\langle 1| - |1\rangle\langle 0|). \end{aligned}$$